

Table 3: Best practices for implementing AIOps

Best practice	Description	Key benefits
Start with small, high-impact use cases	Begin your AIOps journey with narrowly scoped problems—like log noise reduction or predictive alerting—where quick wins are visible and measurable.	Faster ROI; Easier to evaluate success; Builds confidence across teams
Build observability into your pipeline	Ensure your infrastructure, apps, and services emit actionable telemetry—logs, metrics, traces—for AIOps systems to analyse.	Enables accurate anomaly detection; Improves root cause analysis; Strengthens reliability
Keep humans in the loop (HITL) for control	Combine automation with human oversight to review, approve, or override actions taken by AIOps systems, especially in critical environments.	Prevents over-automation errors; Increases trust in AI decisions; Enhances explainability

functionality involved with metrics, logs and traces into one interface. It empowers users to utilise a dashboard-driven view of diagnostics and alerting with CI/CD integrations, cloud services, and an expanding Kubernetes-related risk environment. These solutions are geared towards organisations that need scalable and plug-and-play AIOps.

Open source AIOps platforms

For organisations interested in flexibility, customisation, and cost control, open source AIOps alternatives hold promise, especially when incorporated with existing observability tools. Popular examples include:

ELK Stack (Elasticsearch, Logstash, Kibana) + ML plug-ins: This is a widely adopted set of tools used to analyse logs and metrics. By layering in machine learning extensions, ELK can identify anomalies and trigger real-time alerts.

Prometheus + Cortex + Thanos: Prometheus is mostly used for monitoring, but with the extensions of Cortex and Thanos, it can gain

scalability and increase long-term storage. Machine learning would need to be layered on top of existing tools by using Python, Grafana plugins, or external AI modules.

Apache Spot and OpenAI's **Traces**, along with custom ML models, offer a big-picture view of events and provide predictive diagnostics.

There are some hybrid options too. Various teams use commercial AIOps tools as a layer on top of open source monitoring tools (such as Grafana and Prometheus), making the best of both worlds.

Observability and artificial intelligence are part of every AIOps platform. Observability ensures that systems produce the signals required—metrics, logs, traces, events—while AI produces the insights and acts on these signals. This leads to proactive operations or 'ProOps', the result

of which is faster time to incident response, reduced downtime, smarter use of resources, and more reliable infrastructure.

The integration of artificial intelligence and DevOps signifies an important change in the way modern IT systems are built, managed, and evolved. As we have discussed here, AIOps is not just an extension of a type of automation — it is changing the way operations are modelled from reactive to intelligent, self-healing ecosystems.

That said, building self-healing systems is not just a matter of using new tools — it is about shifting to a new way of thinking that puts a premium on observability, fosters collaboration between humans and AI, and evolves through feedback loops. It means starting small, picking the right use cases, and keeping the human in the loop for trust and governance. **END** 

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